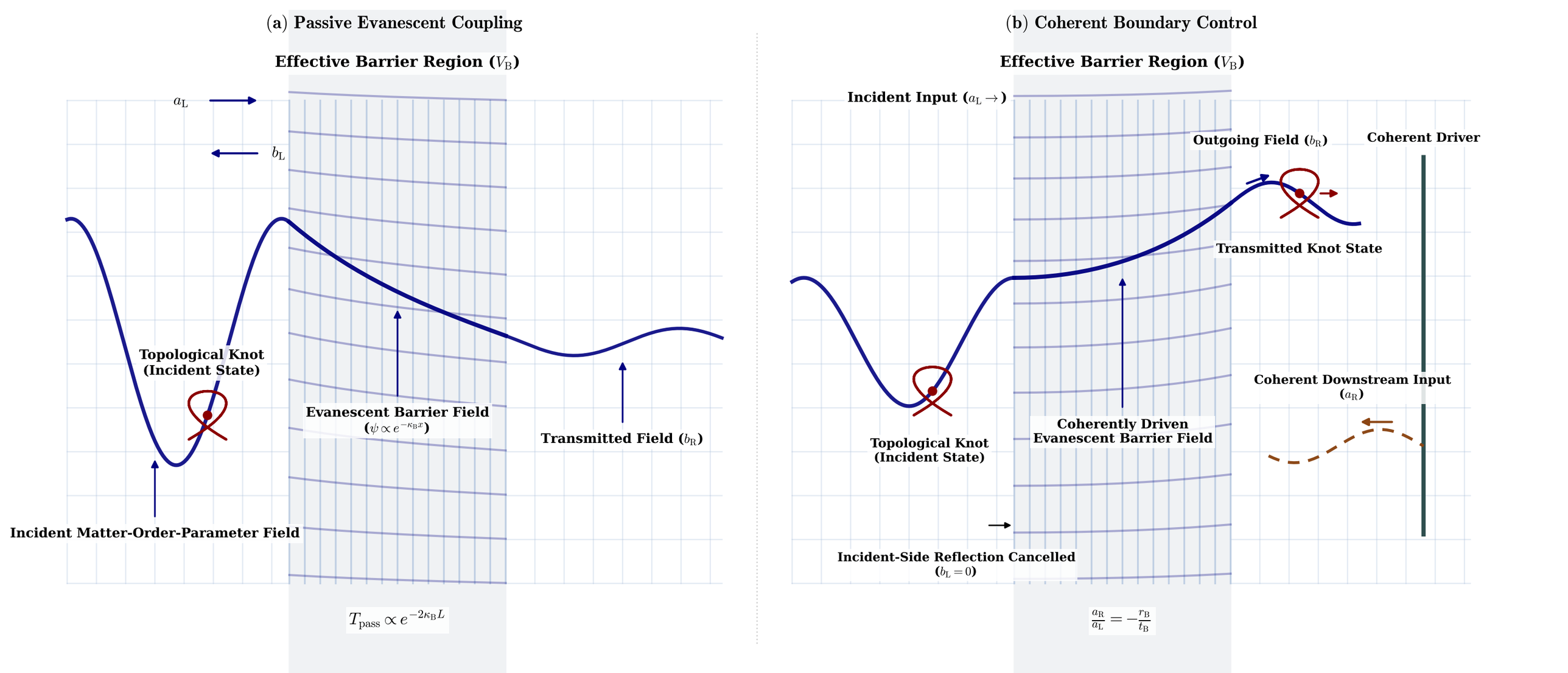

(a) Passive Evanescent Coupling
Effective Barrier Region (V_B)
a_L
b_L
Topological Knot
(Incident State)
Evanescent Barrier Field
($\psi \propto e^{-\kappa_B x}$)
Transmitted Field (b_R)
Incident Matter-Order-Parameter Field
$T_{pass} \propto e^{-2\kappa_B L}$
(b) Coherent Boundary Control
Effective Barrier Region (V_B)
Incident Input ($a_L \rightarrow$)
Outgoing Field (b_R)
Coherent Driver
Transmitted Knot State
Coherent Downstream Input
(a_R)
Topological Knot
(Incident State)
Coherently Driven
Evanescent Barrier Field
Incident-Side Reflection Cancelled
($b_L = 0$)
$\frac{a_R}{a_L} = -\frac{r_B}{t_B}$